Python for Artificial Intelligence

EDA Project: Student Performance Dataset

Week 3, Day 5

Today's Agenda

- Today's whole lesson runs on one real dataset: Kaggle's "Students Performance in Exams"
- Hour 1: Load the data, get your first look, and plan your approach
- Hour 2: Clean and transform it; examine distributions, outliers, and correlations
- Hour 3: Compare groups, combine variables, visualize findings, and present
- By the end: a finished, presentable EDA notebook on a real dataset, your Month 1 portfolio piece

Hour 1

Load, Inspect & Plan

CONCEPT

What Is Exploratory Data Analysis?

- EDA is the process of understanding a dataset before you try to model or draw conclusions from it
- It answers three questions: what does this data actually contain, is it clean, and what patterns stand out
- Every tool from this week, NumPy, Pandas, Matplotlib, and Seaborn, comes together in one EDA project
- A good EDA doesn't just produce charts, it produces a short, clear written story about the data

PROJECT

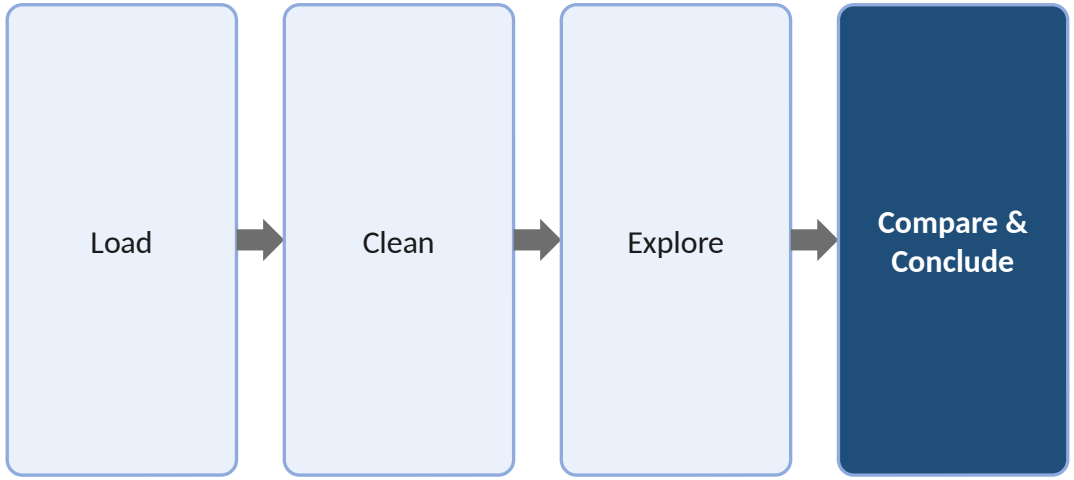
Today's Project: Students Performance in Exams

- Dataset: "Students Performance in Exams" on Kaggle, 1,000 students, 8 columns
- Categorical columns: gender, race/ethnicity, parental level of education, lunch type, test preparation course
- Numeric columns: math score, reading score, writing score (each out of 100)
- The big question: do test preparation, parental education, or lunch type relate to how students score?

CONCEPT

The EDA Workflow

- Step 1: Load the data and take a first look (.head, .info, .describe, .shape)
- Step 2: Clean and transform it (column names, missing values, new features)
- Step 3: Explore distributions, outliers, and correlations
- Step 4: Compare groups and combine variables, then write down what it all means



REFERENCE

Univariate: Which Chart to Use?

One column at a time. The right chart depends on whether it's numeric or categorical.

Variable Type	Best Chart	Example Question
Categorical	Bar chart or count plot	How many students completed test prep?
Numerical	Histogram or box plot	What's the distribution of math scores?

REFERENCE

Bivariate: Which Chart to Use?

Two columns together. The right chart depends on whether each one is numeric or categorical.

Variable Pair	Best Chart	Example Question
Numeric vs. Numeric	Scatter plot or regplot	Does reading score predict writing score?
Numeric vs. Categorical	Box plot per category	Do scores differ by test prep course?
Categorical vs. Categorical	Grouped bar chart or crosstab	Does gender relate to lunch type?

REFERENCE

Multivariate: Which Chart to Use?

Three or more columns at once, the kind of comparison Hour 3 builds up to.

Situation	Best Chart	Example Question
Many numeric columns at once	Correlation heatmap	How do math, reading, and writing relate?
A relationship split by category	Box or scatter plot, colored/grouped by category	Does the test-prep score gap differ by gender?
Many variable pairs at once	sns.pairplot()	Which variable pairs are worth a closer look?

EXAMPLE

Loading and First Inspection

The same four calls from Pandas I, now run on today's actual dataset.

```
import pandas as pd

df = pd.read_csv("StudentsPerformance.csv")

print(df.shape)
print(df.info())
print(df.describe())
print(df.head())
# 1000 rows, 8 columns: 5 categorical, 3 numeric score columns
```

CONCEPT

Planning Your EDA

- Before writing more code, write down 3 to 5 questions this dataset should answer
- Good questions are specific: not "do scores vary" but "do students who complete test prep score higher"
- This dataset's split: numeric (math_score, reading_score, writing_score), categorical (gender, race/ethnicity, parental level of education, lunch, test preparation course)
- Note anything that already looks worth checking: column names with spaces or slashes, any missing values

PROJECT CHECKPOINT

Practice

Download "Students Performance in Exams" from Kaggle and load it. Run .shape, .info(), .describe(), and .head(). Then write down 3 specific questions you want your EDA to answer, and list which columns are numeric versus categorical.

EXAMPLE

What a Good Plan Looks Like

- Example questions for this dataset: "Do students who complete test prep score higher?" or "Does parental education level relate to scores?"
- Another good one: "Is there a gender gap, and does it look the same across all three subjects?"
- Example early observation: "Column names have spaces and slashes, like 'parental level of education', worth renaming"
- This plan doesn't need to be perfect, it just needs to point your next two hours somewhere specific

Hour 2

Clean, Transform & Explore

CONCEPT

Cleaning Checklist

- Rename columns to clean, consistent snake_case: "parental level of education" becomes parental_education
- Check for missing values with df.isnull().sum(), this dataset happens to be clean, but always verify, never assume
- Check data types: all three score columns should already be numeric (int64)
- Create at least one new feature that helps answer your planning questions

EXAMPLE

Cleaning and Feature Engineering

Renaming columns, confirming there are no gaps, and building one new feature.

```
df.columns = ["gender", "race_ethnicity", "parental_education",
              "lunch", "test_prep", "math_score",
              "reading_score", "writing_score"]

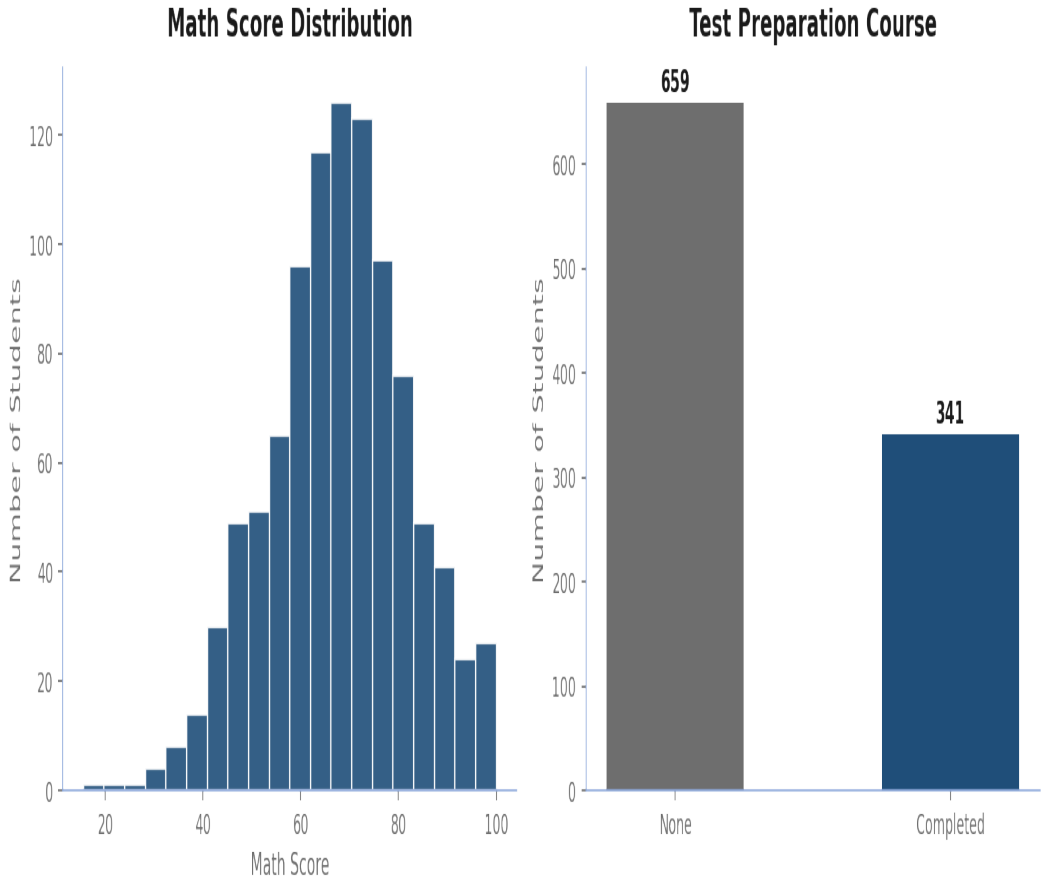
print(df.isnull().sum())
# every column reads 0, nothing missing here

df["average_score"] = df[["math_score", "reading_score",
                          "writing_score"]].mean(axis=1)
```

CONCEPT

Univariate Analysis: Scores and Categories at a Glance

- Start with one column at a time before comparing anything: what does each column look like alone
- A histogram of math_score shows most students cluster in the 50 to 80 range, roughly bell-shaped
- A bar chart of test_prep shows far more students have not completed the course than have
- These single-column views are the baseline every later comparison gets measured against



EXAMPLE

Visualizing Score Distributions

A histogram for the numeric column, a bar chart for the categorical one.

```
import matplotlib.pyplot as plt

plt.hist(df["math_score"], bins=20)
plt.title("Math Score Distribution")
plt.xlabel("Math Score")
plt.show()

df["test_prep"].value_counts().plot(kind="bar")
plt.title("Test Preparation Course")
plt.show()
```

CONCEPT

Distributions & Outliers

- A box plot of average_score shows its shape at a glance, and flags unusually low or high scorers
- An outlier isn't automatically wrong, it might be a genuinely struggling or excelling student
- Decide, and write down, how you're handling each outlier: keep it, flag it, or investigate it, and why
- Never remove an outlier silently, your future self (or a reviewer) needs to know it happened

EXAMPLE

Checking Average Score for Outliers

A box plot shows the shape at a glance; quantile() finds the extreme rows directly.

```
plt.boxplot(df["average_score"])
plt.title("Average Score Distribution")
plt.show()
# a handful of points below the whisker are worth a closer look

bottom_5_percent = df["average_score"].quantile(0.05)
print(df[df["average_score"] < bottom_5_percent])
```

PROJECT CHECKPOINT

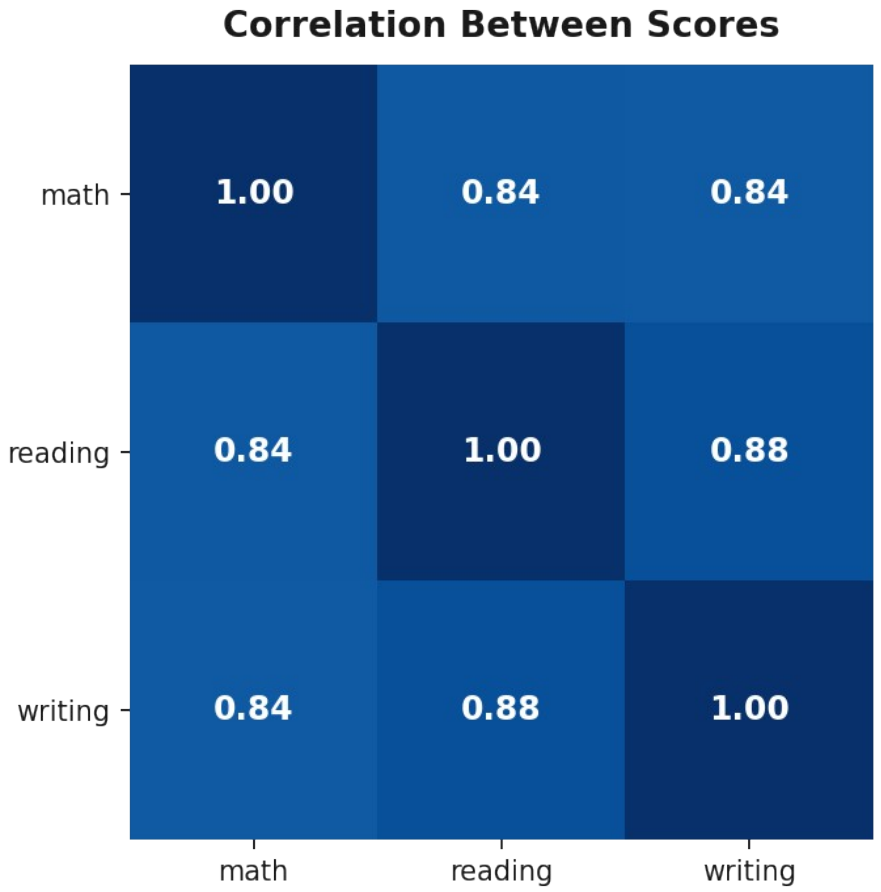
Practice

Plot a histogram of reading_score and a box plot of writing_score. Identify any outliers using quantile(), and write one sentence on whether you'd keep, flag, or investigate them.

CONCEPT

Correlations Worth a Second Look

- Run df[["math_score", "reading_score", "writing_score"]].corr() and scan the numbers
- All three scores usually move together: a student strong in one subject tends to be strong in the others
- A strong correlation is a candidate for a heatmap later, not a guaranteed finding on its own
- This is a first look at a multivariate question, Hour 3 comes back to it in full



PROJECT CHECKPOINT

Practice (Harder)

Clean the column names, check for missing values, create the average_score feature, then check for outliers in average_score and compute the correlation matrix across math_score, reading_score, and writing_score. Note anything above 0.7.

15-Minute Break

Stretch, grab a coffee. We'll resume shortly.

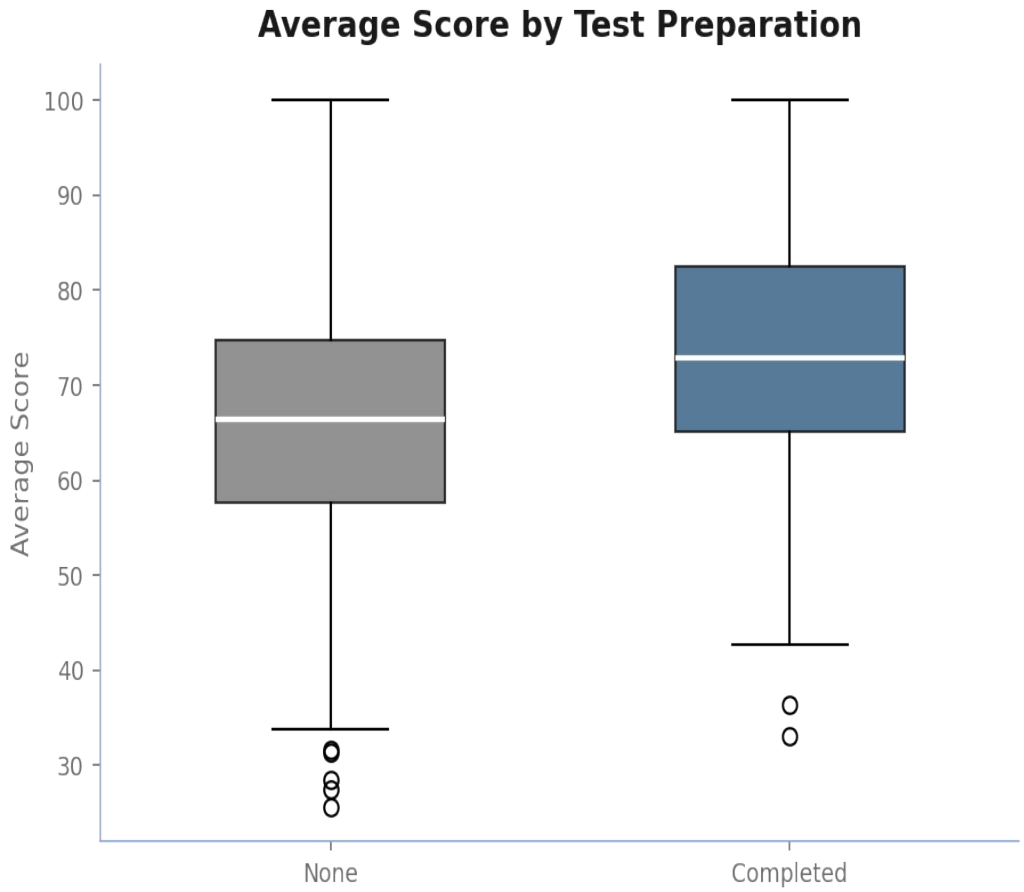
Hour 3

Compare, Conclude & Present

CONCEPT

Bivariate Analysis: Does Test Prep Make a Difference?

- Pair a numeric column (average_score) with a categorical one (test_prep) using a box plot per group
- Students who completed the course tend to sit noticeably higher, with a higher median and a tighter spread
- A box plot side by side is often clearer than comparing two separate histograms
- This is exactly the "numeric vs. categorical" row from the Bivariate reference table in Hour 1



EXAMPLE

Comparing Scores by Test Preparation

A box plot per group, plus the group means behind it.

```
import seaborn as sns

sns.boxplot(x="test_prep", y="average_score", data=df)
plt.title("Average Score by Test Preparation")
plt.show()

print(df.groupby("test_prep")["average_score"].mean())
# completed scores several points higher on average than none
```

CONCEPT

Bivariate Analysis: Reading vs. Writing Score

- Two numeric columns call for a scatter plot or regplot, with .corr() giving the strength as a number
- Reading and writing scores track each other closely, the points hug a clear upward line
- This is a strong, positive relationship, close to the top of the correlation scale from Day 14
- Students strong in reading comprehension tend to also score well in writing, which makes intuitive sense



EXAMPLE

Plotting the Reading-Writing Relationship

A regplot adds the trend line automatically; corr() gives the number behind it.

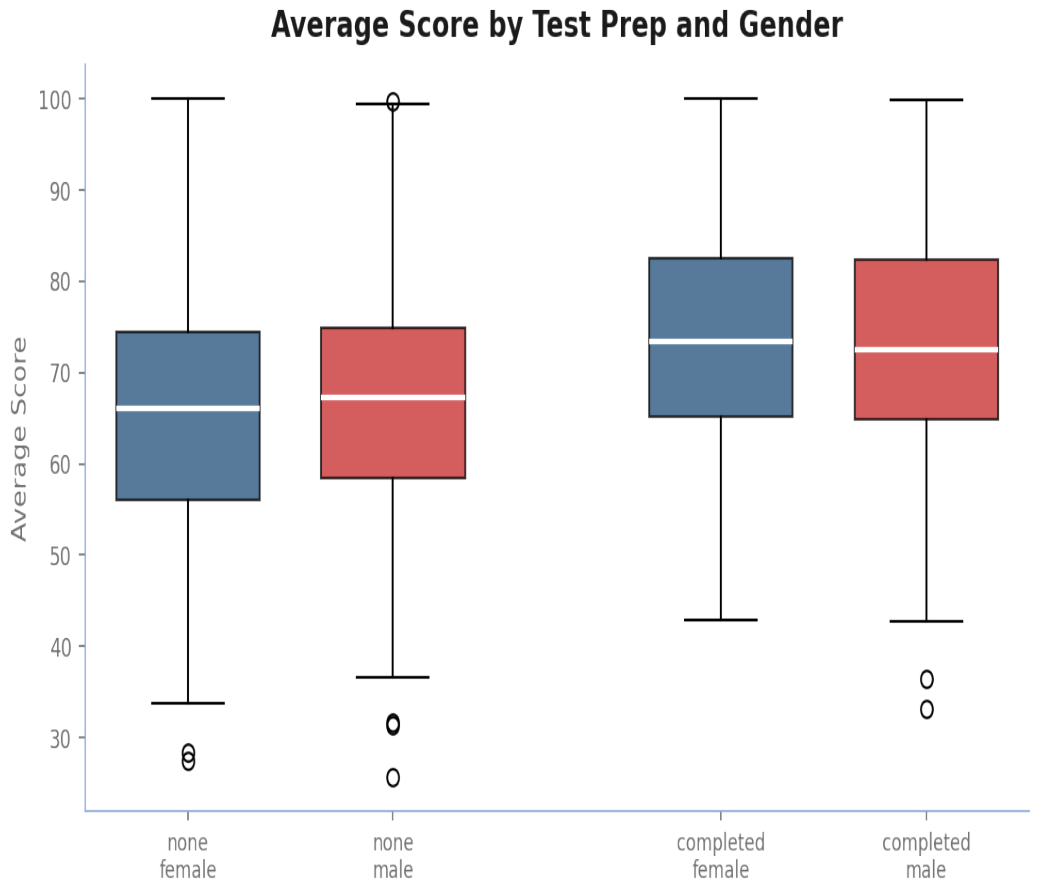
```
sns.regplot(x="reading_score", y="writing_score", data=df,
            scatter_kws={"alpha": 0.5})
plt.title("Reading Score vs. Writing Score")
plt.show()

print(df["reading_score"].corr(df["writing_score"]))
# a strong positive correlation, well above 0.7
```

CONCEPT

Multivariate Analysis: Does the Gap Differ by Gender?

- Adding a third variable means splitting a bivariate chart by one more category
- Here, average_score by test_prep, colored by gender, turns one box plot into four
- The test-prep gap shows up for both genders, roughly the same size either way
- This is how you check whether a pattern holds up everywhere, or only in part of the data



EXAMPLE

A Three-Variable Box Plot

hue= adds the third variable to an otherwise ordinary box plot.

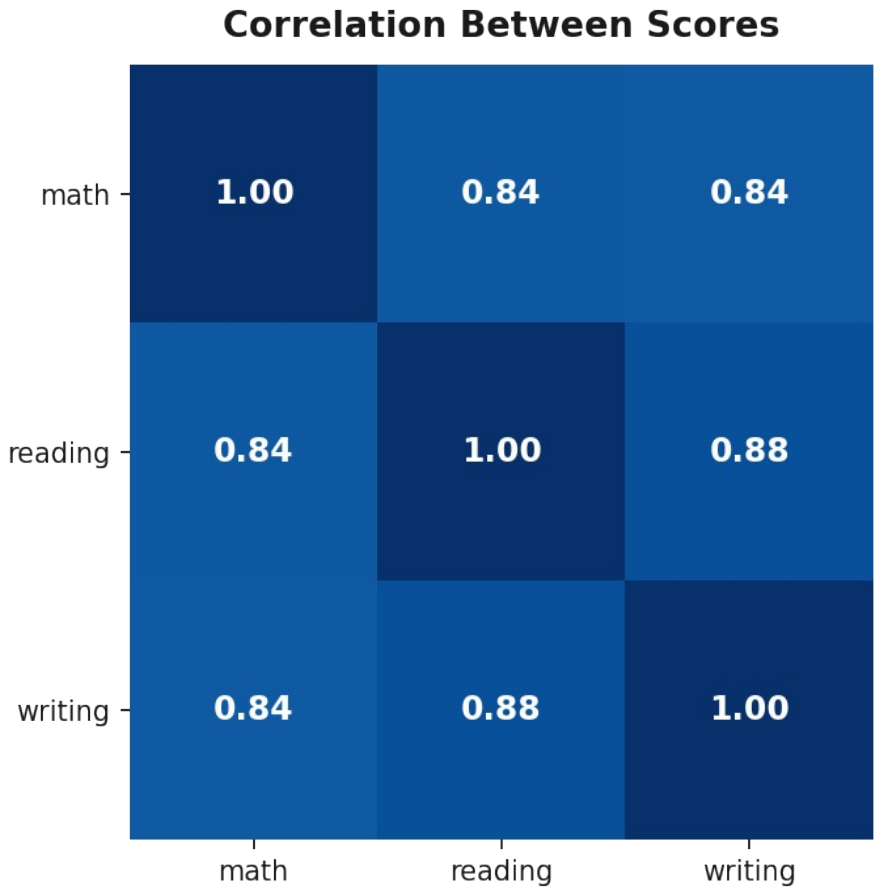
```
sns.boxplot(x="test_prep", y="average_score", hue="gender",
            data=df)
plt.title("Average Score by Test Prep and Gender")
plt.show()

print(df.groupby(["test_prep", "gender"])["average_score"].mean())
```

CONCEPT

Multivariate Analysis: All Three Scores Together

- A full correlation heatmap scans every pair of numeric columns in one picture
- Math, reading, and writing all correlate strongly with each other, none stands alone
- This confirms what the groupby numbers hinted at: overall ability, not just one subject, drives scores
- A heatmap is often the single most useful multivariate chart to start with on a new dataset



EXAMPLE

A Full Correlation Heatmap

annot=True prints each correlation value directly on the heatmap.

```
corr_matrix = df[["math_score", "reading_score",
                  "writing_score"]].corr()

sns.heatmap(corr_matrix, annot=True, cmap="Blues")
plt.title("Correlation Between Scores")
plt.show()
```

PROJECT CHECKPOINT

Practice

Pick one more grouping of your own: parental_education vs. average_score, or lunch vs. average_score. Make a bivariate chart for it, then add a third variable (gender or race_ethnicity) to turn it into a multivariate chart.

CONCEPT

Writing Your Conclusions

- For each chart, write 2 to 3 sentences: what does it show, and why does it matter
- State findings in plain language: "students who completed test prep scored about 7 points higher on average", not "see chart"
- Be honest about limitations: this is self-reported survey data, it doesn't prove test prep caused the gap
- Three clear, well-supported findings beat ten vague, unsupported ones

PROJECT CHECKPOINT

Practice

Finalize 2 to 3 charts from today that each answer one of your planning questions. For each, write 2 to 3 sentences stating the finding in plain language, plus one limitation of your analysis.

CONCEPT

Presenting Your Findings

- You have about 3 minutes: state the dataset and question, show your best chart, state the finding
- Skip the code walkthrough, the audience wants your conclusion, not your syntax
- End with one honest limitation, this shows judgment, not weakness
- This project goes straight into your portfolio as your first real, presentable data analysis

MILESTONE

Month 1 Milestone Checklist

- Loaded the Students Performance dataset and ran a first inspection: .shape, .info(), .describe(), .head()
- Wrote 3 to 5 specific planning questions before exploring
- Cleaned the column names, confirmed there were no missing values, and created average_score
- Checked for outliers in average_score and computed a full correlation matrix across all three scores
- Compared groups with at least one bivariate chart and one multivariate chart
- Produced 2 to 3 labeled charts, each with a plain-language conclusion

Key Takeaways & Coming Up

- EDA turns a raw dataset into a short, clear story: what's here, is it clean, and what patterns matter
- On this dataset: test preparation and gender both relate to scores, and all three subjects move together
- Choosing the right chart for univariate, bivariate, and multivariate questions is a repeatable skill, not guesswork
- This project is your Month 1 portfolio piece, the first of several you'll build in this course
- Coming up next: Month 2 begins with classical Machine Learning, starting with the ML workflow and Linear Regression